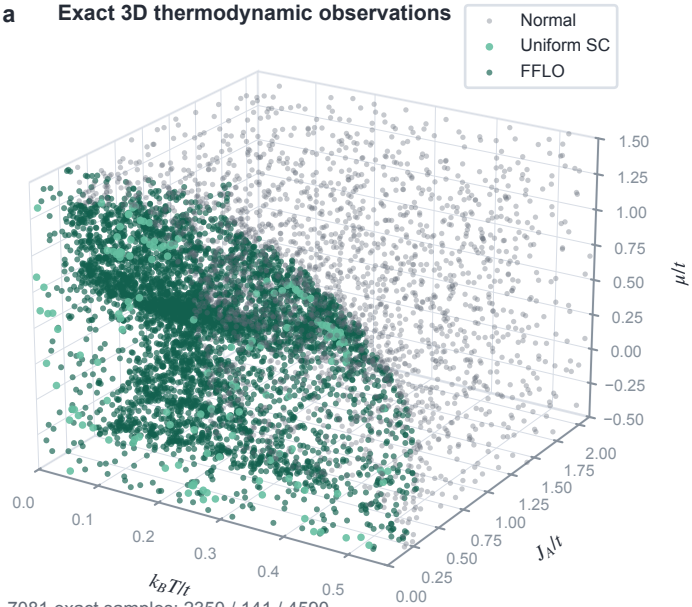
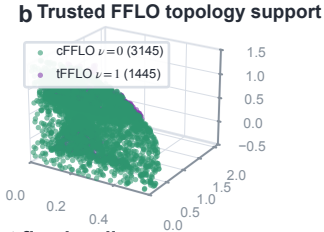
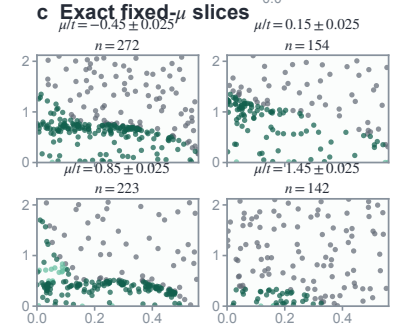
Reliability-aware active learning extends to higher-dimensional exploration

3D extension reveals thermodynamic and topology-aware structures; strict topology closure remains open







d Strict 3D topology gates: not converged

FAIL	map change	0.002561 > 0.002
FAIL	surface shift p95	0.005996 > 0.004167
FAIL	coverage p95	0.015318 > 0.00625
FAIL	trusted surprise	0.140625 > 0.02

components $3 \to 4 \to 3$; no surface; 4/20 slices
limited domain: $\mu/t \in [-0.5, 1.5]$